



AIDING PLANT GROWTH

While asteroids have had a history of having destructive effects upon landing on Earth's surface, recent studies have made many rethink this notion of asteroids being destroyers. A recent study, published in July this year, suggested that we could soon see astronauts consuming salads that were grown using extraterrestrial soil. The study titled CI Asteroid Regolith as an In Situ Plant Growth Medium for Space Crop Production, published by The American Astronomical Society contained enough evidence to suggest that it was indeed possible to grow crops as we do here on Earth, in space too! This breakthrough is important because if we are looking to go ahead and settle on other planets, then providing adequate nutrition to the populace will be an important thing to consider while drafting the policies. Right now, the primary method to feed astronauts is through pre-packaged meals. While this may be enough for missions that last a few weeks at max, in the long run, we will have to devise ways in which we can produce enough food in the place where the travellers are headed to. And, this discovery could play a pivotal role in taking us in the right direction.

The researchers in the paper noted, "Right now, our current method of feeding astronauts

consists of resupplied prepackaged food from Earth, which is unsustainable for long-term missions. Using planetary resources via in situ resource utilization to grow crops is the next step toward sustainability in space. Asteroids are an abundant space resource and should not be overlooked when considering crewed missions. In particular, the primordial CI carbonaceous asteroids are of interest



because the regolith is suggested to contain soluble elemental nutrients, such as phosphorous and potassium, that crops can use for growth and development."

The results of the study give positive indications that the stimulant present the studied sample of the CI Asteroid Regolith is enough to sustain plant life, with certain

limitations. These samples had been collected from the Hayabusa2 spacecraft upon its return to Earth. However, further investigation into the matter is necessary to determine the mitigation strategies to ensure that the regolith in space can sustain plant life in the long run.

LOOKING BACK IN TIME

Well, if there was still a doubt lingering in your mind about the utility of information that asteroids and their related materials can provide us, then this should seal the deal for you. The samples collected from Hayabusa2, when it came back to Earth. Upon its landing in Australia, researchers for the first time ever, got their hands on asteroid Earth. And, when it was studied further by the folks at Japan Aerospace Exploration Agency.

Hayabusa2 had been exploring Ruyugu for a long time before it came back to Earth, bearing two smidgens of dirt from the asteroid. It was an ancient, carbon-rich asteroid, and its texture has been described to have the "texture of freeze-dried coffee". These details

are not as relevant compared to what the researchers found out after conducting their studies.

The study of the 1,534 grains of asteroid material that had been brought back to Earth has been crucial in understanding how the asteroids were formed and the way in which they hold themselves together. **d**